

Inverse theorems for MDS Codes with small Schur squares

Alain Couvreur, **Nihan Tanısalı**, Gilles Zémor

26 June 2026

ENCODE Workshop 2026



Inverse Theorems in Additive Combinatorics

An inverse theorem derives structural information from quantitative additive constraints.

Vosper Theorem

Let $A, B \subset \mathbb{Z}/p\mathbb{Z}$ with $|A|, |B| \geq 2$. If

$$|A + B| = |A| + |B| - 1 \leq p - 2$$

where

$$A + B := \{a + b \mid a \in A, \quad b \in B\},$$

then A and B are arithmetic progressions with the same common difference.

- The objective is to build coding theory analogs of Vosper-like statements.

Plan of the Talk

- 1 Introduction
- 2 Multiplicative Analog of Vosper Theorem
- 3 $+1$ Regime
- 4 Partial Answers

Schur Product

Definition

- The Schur product of $\alpha = (\alpha_1, \dots, \alpha_n), \beta = (\beta_1, \dots, \beta_n) \in \mathbb{F}_q^n$ is defined as

$$\alpha \star \beta := (\alpha_1 \beta_1, \dots, \alpha_n \beta_n)$$

- Given two linear codes $\mathcal{A}, \mathcal{B} \subset \mathbb{F}_q^n$, the Schur product

$$\mathcal{A} \star \mathcal{B} := \langle \alpha \star \beta \mid \alpha \in \mathcal{A} \text{ and } \beta \in \mathcal{B} \rangle$$

- For a linear code $\mathcal{C}$, the Schur product $\mathcal{C} \star \mathcal{C}$ (or $\mathcal{C}^2$) is called the square of $\mathcal{C}$

Generalised Reed-Solomon Code

Let $\alpha = (\alpha_1, \dots, \alpha_n)$ be an n -tuple of pairwise distinct elements of $\mathbb{F}_q$ and $\vec{v} \in (\mathbb{F}_q \setminus \{0\})^n$. Generalised Reed-Solomon code of dimension k is defined as

$$\text{GRS}_k(\vec{\alpha}, \vec{v}) := \{(v_1 f(\alpha_1), \dots, v_n f(\alpha_n)) \mid f \in \mathbb{F}_q[x]_{<k}\}.$$

What is Already Known?

Vosper Theorem

Let $A, B \subset \mathbb{Z}/p\mathbb{Z}$ with $|A|, |B| \geq 2$. If

$$|A + B| = |A| + |B| - 1 \leq p - 2$$

where

$$A + B := \{a + b \mid a \in A, \quad b \in B\},$$

then A and B are arithmetic progressions with the same common difference.

Theorem- Coding Theory Analogue of Vosper (D. Mirandola and G. Zémor)

Let $\mathcal{C}, \mathcal{D} \in \mathbb{F}_q^n$ be MDS codes with dimension ≥ 2 . Assume that

$$\dim \mathcal{C} \star \mathcal{D} = \dim \mathcal{C} + \dim \mathcal{D} - 1 \leq n - 2.$$

Then $\mathcal{C}$ and $\mathcal{D}$ admits a basis in geometric progression with the same difference vector. That is to say

$$\mathcal{C} = \langle g, g \star \alpha, \dots, g \star \alpha^{k-1} \rangle \quad \text{and} \quad \mathcal{D} = \langle h, h \star \alpha, \dots, h \star \alpha^{\ell-1} \rangle.$$

Analogy (Additive Combinatorics $\longrightarrow$ Coding Theory)

- ▶ Subset $\longrightarrow$ Code
- ▶ Addition $\longrightarrow$ Schur product
- ▶ Cardinality $\longrightarrow$ Dimension
- ▶ Arithmetic progression $\longrightarrow$ A basis in geometric progression

$$\{0, 1, \dots, k-1\} \rightarrow \langle g \star 1, g \star \alpha, \dots, g \star \alpha^{k-1} \rangle$$

MDS Condition

An Extreme Example

Let $\mathcal{C}, \mathcal{D} \subset \mathbb{F}^4$ be generated by

$$G_{\mathcal{C}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad G_{\mathcal{D}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then $\mathcal{C}$ and $\mathcal{D}$ have disjoint supports. Therefore, we have

$$\mathcal{C} \star \mathcal{D} = \{0\}.$$

MDS Condition

An Extreme Example

Let $\mathcal{C}, \mathcal{D} \subset \mathbb{F}^4$ be generated by

$$G_{\mathcal{C}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad G_{\mathcal{D}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then $\mathcal{C}$ and $\mathcal{D}$ have disjoint supports. Therefore, we have

$$\mathcal{C} \star \mathcal{D} = \{0\}.$$

Theorem (Randriambololona)

Let $\mathcal{C}, \mathcal{D} \subset F^n$ be full support codes. If at least one of them is MDS, then

$$\dim \mathcal{C} \star \mathcal{D} \geq \min(n, \dim \mathcal{C} + \dim \mathcal{D} - 1).$$

► $\dim \mathcal{C} \star \mathcal{D} = \dim \mathcal{C} + \dim \mathcal{D} - 1 + \gamma$ where γ is a small integer.



Randriambololona, Hugues (2015). "On products and powers of linear codes under componentwise multiplication". In: *Algorithmic Arithmetic, Geometry and Coding Theory*. Vol. 637. Contemp. Math. Amer. Math. Soc.

- 1 Introduction
- 2 Multiplicative Analog of Vosper Theorem
- 3 $+1$ Regime
- 4 Partial Answers

An Overview

	$\gamma = 0$	$\gamma = 1$
Additive Case	Vosper $ A + B = A + B - 1$ $\implies A = \{a, a + d, \dots, a + (k - 1)d\}$ $B = \{b, b + d, \dots, b + (\ell - 1)d\}$	Hamidoune–Rødseth Hamidoune–Serra–Zémor $ A + B = A + B $ $\implies A \subseteq \{a, a + d, \dots, a + (k - 1)d\}$ $B \subseteq \{b, b + d, \dots, b + (\ell - 1)d\}$ with one missing element
Multiplicative Case	Mirandola–Zémor $\dim \mathcal{C} \star \mathcal{D} = \dim \mathcal{C} + \dim \mathcal{D} - 1$ $\implies \mathcal{C} = \langle g, g \star \alpha, \dots, g \star \alpha^{k-1} \rangle$ $\mathcal{D} = \langle h, h \star \alpha, \dots, h \star \alpha^{\ell-1} \rangle$	Current Question $\dim \mathcal{C} \star \mathcal{D} = \dim \mathcal{C} + \dim \mathcal{D}$ $\implies ?$

-  Hamidoune, Yahya Ould and Oystein J. Rodseth (2000). “An inverse theorem mod p ”. In: *Acta Arithmetica* 92.3, pp. 251–262.
-  Hamidoune, Yahya Ould, Oriol Serra, and Gilles Zémor (2008). “On the critical pair theory in abelian groups: Beyond Chowla’s Theorem”. In: *Combinatorica* 28.4.
-  Mirandola, Diego and Gilles Zémor (2015). “Critical pairs for the Product Singleton Bound”. In: *IEEE Trans. Inform. Theory* 61.9, pp. 4928–4937.

- We consider only the symmetric case $\mathcal{D} = \mathcal{C}$.

The sets satisfying $\gamma = 1$ are precisely of the form

$$A_1 = \{0, \widehat{1}, 2, 3, \dots, k\} \quad \text{and} \quad A_2 = \{0, 1, 2, \dots, \widehat{k-1}, k\}.$$

Multiplicative analogs of A_1 and A_2 are

$$\mathcal{C}_1 = \langle 1, \widehat{\alpha}, \alpha^2, \dots, \alpha^k \rangle \quad \text{and} \quad \mathcal{C}_2 = \langle 1, \alpha, \alpha^2, \dots, \widehat{\alpha^{k-1}}, \alpha^k \rangle.$$

We have the following sumsets and Schur squares:

- $A_1 + A_1 = \{0, \widehat{1}, 2, \dots, 2k\}$ and $A_2 + A_2 = \{0, 1, \dots, \widehat{2k-1}, 2k\},$
- $\mathcal{C}_1 \star \mathcal{C}_1 = \{1, \widehat{\alpha}, \alpha^3, \dots, \alpha^{2k}\}$ and $\mathcal{C}_2 \star \mathcal{C}_2 = \{\alpha^1, \alpha, \dots, \widehat{\alpha^{2k-1}}, \alpha^{2k}\}.$

- The codes $\mathcal{C}_i$ for $i = 1, 2$ satisfy the following $\dim \mathcal{C}_i^2 = 2 \dim \mathcal{C}_i$
- **Question:** Are they the only codes attaining this equality?

Answer: No

Theorem (Mumford)

Let G, G' be two divisors of a function field F of genus g over an algebraically closed field K . Suppose that $\deg G > 2g$ and $\deg G' > 2g + 1$. Then

$$\mathcal{L}(G)\mathcal{L}(G') = \mathcal{L}(G + G').$$

Algebraic Geometric Codes of Genus 1

Let $\mathcal{E}/\mathbb{F}_q$ be an elliptic curve with rational points $P_1, \dots, P_n \in \mathcal{E}(\mathbb{F}_q)$, and let

$$D := P_1 + \dots + P_n.$$

Let G be a divisor with $G \cap D = \emptyset$. The *elliptic code* associated to (D, G) is

$$C_{\mathcal{L}}(D, G) := \{ (f(P_1), \dots, f(P_n)) \in \mathbb{F}_q^n \mid f \in \mathcal{L}(G) \}.$$

► If $\deg G, \deg G' \geq 2$, then by Riemann–Roch theorem, we have

$$C_{\mathcal{L}}(D, G) \star C_{\mathcal{L}}(D, G') = C_{\mathcal{L}}(D, G + G')$$

1 Introduction

2 Multiplicative Analog of Vosper Theorem

3 $+1$ Regime

4 Partial Answers

Two Subcases

Let $\dim \mathcal{C} = k$. We assume

$$\dim \mathcal{C}^2 = 2 \dim \mathcal{C}.$$

Therefore $\mathcal{C}^2$ is MDS if

$$d_{\min}(\mathcal{C}^2) = n - \dim \mathcal{C}^2 + 1 = n - 2k + 1.$$

We give results where

- ▶ $d_{\min}(\mathcal{C}^2) \leq n - 2k - 1$
- ▶ $d_{\min}(\mathcal{C}^2) = n - 2k + 1$ (MDS)

MDS –2 Schur Square

Theorem (Couvreur, T., Zémor)

Let $\mathcal{C}$ be an MDS code with $\dim \mathcal{C} = k \geq 2$. Assume that

$$\dim \mathcal{C}^2 = 2 \dim \mathcal{C} = 2k, \quad \text{and} \quad d = d_{\min}(\mathcal{C}^2) \leq n - 2k - 1.$$

Then there is a set I with $|I| = n - d$ such that $\text{punc}(\mathcal{C}, I)$ is a Reed–Solomon code.

- Liu, Luo, Wang, and Zhai consider the code

$$\mathcal{C} = \{(f(\alpha_1), \dots, f(\alpha_{n-1}), f(\alpha_n) - \lambda f(\alpha_{n+1})) : \deg f < k\}$$

with $\alpha_1, \dots, \alpha_{n-1}, \alpha_n, \alpha_{n+1}, \lambda \in \mathbb{F}_q$.

- Under some mild conditions on λ , the code $\mathcal{C}$ is MDS and

$$\dim \mathcal{C}^2 = 2 \dim \mathcal{C} = 2k$$

and $d_{\min}(\mathcal{C}^2) = 1$.

- $\text{punc}_{[1, n-1]}(\mathcal{C})$ is a Reed–Solomon code.

 Liu, Wei, Jinquan Luo, Puyin Wang, and Dengxin Zhai (2025). “Column Twisted Reed–Solomon Codes as MDS Codes”. In: *arXiv preprint arXiv:2507.08755*.

Lemma (Couvreur, T., Zémor)

Let $\mathcal{C} \subseteq F^n$ be a code with $\dim \mathcal{C} \geq 2$ and $d_{\min}(\mathcal{C}) > 1$. Assume there exists a 2 dimensional MDS code $\mathcal{A} = \text{RS}_2(\alpha) \subseteq F^n$ for some $\alpha \in F^n$ distinct entries such that

$$\dim \mathcal{AC} = \dim \mathcal{A} + \dim \mathcal{C}.$$

Then $\mathcal{C}$ is the sum of two Reed–Solomon codes, that is

$$\mathcal{C} = \text{RS}_s(\alpha) + f \star \text{RS}_t(\alpha).$$

$$\begin{pmatrix} \begin{array}{ccccccc} 1 & 1 & \cdots & 1 & 1 & \cdots & 1 \\ \alpha_1 & \alpha_2 & \cdots & \alpha_w & \alpha_{w+1} & \cdots & \alpha_n \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{t-1} & \alpha_2^{t-1} & \cdots & \alpha_w^{t-1} & \alpha_{w+1}^{t-1} & \cdots & \alpha_n^{t-1} \end{array} \\ \begin{array}{ccccccc} f_1 & f_2 & \cdots & f_w & 0 & \cdots & 0 \\ f_1 \alpha_1 & f_2 \alpha_2 & \cdots & f_w \alpha_w & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ f_1 \alpha_1^{s-1} & f_2 \alpha_2^{s-1} & \cdots & f_w \alpha_w^{s-1} & 0 & \cdots & 0 \end{array} \end{pmatrix}$$

MDS Schur Square

Theorem (Couvreur, T., Zémor)

Let $\mathcal{C} \in \mathbb{F}_q^n$ be an MDS code with dimension ≥ 2 . Assume that

$$\dim \mathcal{C} \star \mathcal{C} = 2 \dim \mathcal{C} \leq n - 4,$$

and assume that $\mathcal{C}^2$ is an MDS code. Then $\mathcal{C}$ is the sum of two Reed–Solomon codes, that is

$$\mathcal{C} = \text{RS}_s(\alpha) + f \star \text{RS}_t(\alpha).$$

Examples

Consider an odd integer k .

- Consider our first example $\mathcal{C} = \langle 1, \hat{\alpha}, \alpha^2, \dots, \alpha^{(k-1)} \rangle$. This code can be written as a sum

$$\mathcal{C} = \text{RS}_{(k+1)/2}(\alpha^2) + \alpha^3 \star \text{RS}_{(k-3)/2}(\alpha^2).$$

- Elliptic curves in Weierstrass form give rise to codes

$$\mathcal{C} = \text{RS}_{(k+1)/2}(\alpha) + \beta \star \text{RS}_{(k-1)/2}(\alpha)$$

The Objective

Classify all MDS codes $\mathcal{C}$ with

$$\dim \mathcal{C}^2 = 2 \dim \mathcal{C}$$

and $\mathcal{C}^2$ is MDS.

- ▶ Algebraic geometric codes from elliptic curves
- ▶ Codimension one subcodes of Reed–Solomon codes
- ▶ Something else?

The Objective

Classify all MDS codes $\mathcal{C}$ with

$$\dim \mathcal{C}^2 = 2 \dim \mathcal{C}$$

and $\mathcal{C}^2$ is MDS.

- ▶ Algebraic geometric codes from elliptic curves
- ▶ Codimension one subcodes of Reed–Solomon codes
- ▶ Something else?

Thank You For Your Attention!